

Usability Study Report

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Application: FieldTrip (<https://fieldtrip.onrender.com>)

Course: CS 5610, Northeastern University

Recordings: <https://drive.google.com/drive/folders/1urwTCxpXr-SIFP9Y1-3awzjychyXsbjs?usp=sharing>

Application scope

Application description:

FieldTrip is a web app I built with my teammate for CS 5610. It lets college students post activities they've discovered (hiking spots, cheap food, sports, culture, nightlife), plan group trips around those activities, and leave ratings and reviews afterward. The goal is to make it easier to find things to do and to know which organizers are actually reliable.

Users, target audience:

- Maya, the explorer. 21, junior. Always hunting for new things to do, browses by category and difficulty, joins trips on impulse.
- Jake, the organizer. 22, senior. Plans weekend hikes for his friend group, wants to manage RSVPs and know who's actually showing up.
- Priya, the reviewer. 20, sophomore. Won't join a trip unless the organizer has good ratings, reads feedback carefully, always leaves detailed reviews.

Data description:

Three main collections. Activities (title, category, difficulty, cost, group size, location, format, description). Trips (activity reference, organizer, date, meeting point, RSVP list, status, feedback). Users (username, bio, rating, trips organized and attended).

Main tasks, use cases:

- T1: Find an outdoor activity under 10 dollars using search and filters.
- T2: Create an account and post a new activity you discovered.
- T3: Schedule a group trip and join a trip someone else organized.
- T4: Open My Trips, find a trip you organized, update the meeting point.
- T5: Check personal stats and update your profile bio.

Experiment

Preparation

Introduction

- Welcome participant
- Ask for consent to record
- Ask to think out loud
- Remind them they aren't being evaluated, can leave at any time
- Ask participant to think like a college student looking for things to do

Demographics questions (could decline any):

- Age
- Gender (optional)
- Year in school or occupation
- How often they use web apps
- Device and browser

Recording setup:

Zoom with screen share plus audio. Files saved locally, uploaded to a Google Drive folder shared with john.guerra@gmail.com.

Script of tasks read to participants:

- Initial approach: "Look at the site and tell me what you think it is, what it does, and what you'd do first."
- T1: "Imagine it's Friday and you want something outdoors for the weekend that doesn't cost much. Try to find some options."
- T2: "You just found a really good cheap spot (taco place, bakery, pizza, whatever). Post it on the app so others can find it."
- T3: "A friend mentioned a hiking trip this weekend. Find it and join. Then schedule your own trip for the place you just posted."
- T4: "A day later the plan changes. Update the meeting point for the trip you organized."
- T5: "Find where you can see your own activity and update your profile bio."

Post questionnaire Likert scales (1 to 5):

- How effective was the application for T1
- How intuitive and easy to use was the application for T1
- (Same two questions for T2, T3, T4, T5)
- How effective was the application overall
- How intuitive and easy to use was the application overall
- Any final comments or suggestions for improvement

Experiment Notes

After each session I wrote notes right away, then rewatched the recording to catch moments I missed. The notes below combine both passes.

Participant 1

Demographics answers

- Age: 25
- Gender: Male
- Occupation: CS Major
- Uses web apps: daily
- Device: MacBook Air, Chrome

Detailed notes

- Initial approach: Figured out the app in about 10 seconds. Read the three feature cards and said "find things, plan trips, leave reviews, got it." Clicked browse activities before I gave any prompt.
- T1 (browse and filter): Went straight to filters. Set format to outdoor, cost max to 10, then added hiking. No hesitation. Said the cards were easy to scan.
- T2 (sign up and post): Signed up in under a minute. Posted the activity fast. Mentioned he wished there was a photo upload because "all of these are just text."
- T3 (schedule and join): Clicked into a hiking activity, saw the trips section, joined one, then scheduled his own. Smooth. Said the spot counter updating was a nice touch.
- T4 (manage trips): Found My Trips immediately, saw organized vs joined split, clicked edit, updated meeting point. No issues.
- T5 (stats and profile): Went to My Stats, read the numbers, liked the personal breakdown. Found profile by clicking his name in the nav.

Post-test questionnaire results (Likert scales)

- T1 score: 5 effective, 5 intuitive. Filters did exactly what he wanted and updated fast.
- T2 score: 5 effective, 4 intuitive. Docked a point because of the missing photo option.
- T3 score: 4 effective, 4 intuitive. Had to scroll slightly on the activity page to reach the trips section.
- T4 score: 4 effective, 4 intuitive. Clean layout, edit worked first try.
- T5 score: 4 effective, 4 intuitive. Stats are good, profile link being just his name is standard but could be more obvious.
- Overall: 4 effective, 4 intuitive. "Solid project. Add photos and a sort option and it's great."

Participant 2

Demographics answers

- Age: 23
- Gender: Female
- Occupation: Data Science Major
- Uses web apps: daily
- Device: Dell laptop, Chrome

Detailed notes

- Initial approach: Took about 25 seconds to read the landing page. Thought it was a travel app at first, then corrected herself. Asked if she needed to sign up before she could look.
- T1 (browse and filter): Typed “outdoor” in the search bar first. Didn’t notice the filter dropdowns until I waited. Said the filters kind of blend into the background and could use a label or stronger color.
- T2 (sign up and post): Clicked Log In by mistake before Sign Up. Found the switch link at the bottom. Once logged in, she had to scan for the Post Activity button in the nav for a few seconds.
- T3 (schedule and join): Noticed the trips section was kind of low on the activity page. Said she almost missed it. The datetime picker made her pause briefly because she wasn’t sure of the format.
- T4 (manage trips): Found My Trips without issue. Edit Trip Details was clear once she spotted it.
- T5 (stats and profile): Went to My Stats, said the page felt empty since she just signed up. Took a second to realize clicking her name in the nav opened the profile page.

Post-test questionnaire results (Likert scales)

- T1 score: 4 effective, 3 intuitive. Filters worked but she almost missed them. They should stand out more.
- T2 score: 4 effective, 3 intuitive. Login vs sign up flow tripped her up, and the Post Activity button is not prominent enough.
- T3 score: 4 effective, 4 intuitive. Once she saw the trips section it was fine, datetime picker was clearer after she opened it.
- T4 score: 4 effective, 4 intuitive. Clear tabs and edit button.
- T5 score: 4 effective, 3 intuitive. Profile link being just a name isn’t obvious. Stats felt bare for a brand new user.
- Overall: 4 effective, 3 intuitive. “It works, but a few things could stand out more for first time users.”

Participant 3

Demographics answers

- Age: 25
- Gender: Male
- Occupation: Software Engineer
- Uses web apps: daily
- Device: MacBook Pro, Firefox

Detailed notes

- Initial approach: Noticed the dark theme right away. Compared it mentally to Meetup but said the review system was the interesting differentiator. I had to redirect him a couple of times when he slipped into design critique mode.
- T1 (browse and filter): Confident with filters. Combined difficulty, format, and cost in one go. Said he wanted sorting options like “newest” or “most joined” since the listings all looked similar at a glance.
- T2 (sign up and post): Signed up quickly. Posted the activity without issues but flagged the missing image upload as a real gap for a discovery app. Also asked about a character limit on the description.
- T3 (schedule and join): Scheduled a trip fine. Liked that past dates are blocked on the date picker. When joining, he wanted to see who the organizer was, specifically their past trip count and rating, to decide whether to commit.
- T4 (manage trips): Found My Trips easily. Said the inline edit felt a little cramped, wondered if a modal would be cleaner. Task completed fine.
- T5 (stats and profile): Said the stats page felt empty for a brand new user. Suggested an onboarding prompt instead of all zeros. Profile edit worked.

Post-test questionnaire results (Likert scales)

- T1 score: 4 effective, 4 intuitive. Filters worked but he wants sorting and some popularity signal.
- T2 score: 4 effective, 4 intuitive. Clean form but no images and no character limit hint.
- T3 score: 4 effective, 3 intuitive. The missing organizer visibility bugged him most. "I can't tell if this person is reliable."
- T4 score: 4 effective, 4 intuitive. Edit worked. Inline edit slightly squeezed.
- T5 score: 3 effective, 4 intuitive. Empty state for new users was the main issue.
- Overall: 4 effective, 4 intuitive. "Solid for a class project. The review system angle is strong but the trust signals need to be more visible."

Prioritized list of issues and corresponding changes

The most crucial issues that came up across my three sessions, ranked with MoSCoW priority.

Issue 1

Issue: Two participants didn't notice the filter bar on the activities page at first. It blends into the background.

Change: Give the filter section a distinct background or a "Filters" label so it reads as its own area.

Priority: Should

Was it implemented? How? Yes. I bumped the contrast on the filter bar background and added a small header so it stands out from the card grid below.

Issue 2

Issue: Post Activity button in the navbar doesn't stand out. One participant had to scan for it after logging in.

Change: Use the accent color on that link so it reads as a primary action.

Priority: Should

Was it implemented? How? Yes. Post Activity now uses the orange accent instead of the default gray nav style.

Issue 3

Issue: Profile link in the navbar is just the user's name. A participant said a lot of first time users would miss that.

Change: Add a small icon next to the name or use the word "Profile" explicitly.

Priority: Could

Was it implemented? How? Not in this iteration. Decided it was lower priority than the filter and Post Activity fixes.

Issue 4

Issue: No image upload on activities. All three participants brought this up in some way. Listings feel interchangeable without visuals.

Change: Add image upload on the activity form, with category fallback icons.

Priority: Would

Was it implemented? How? Not in this iteration. Image hosting is out of scope for now. Noted as a future enhancement.

Issue 5

Issue: My Stats page feels empty for brand new users. Just zeros and ones, no encouragement to do anything.

Change: Add a friendly empty state for new users with prompts like "post your first activity" or "join a trip".

Priority: Should

Was it implemented? How? Yes. New users now see a welcome panel with shortcut buttons to the activities page and the post form.

Issue 6

Issue: No way to sort activities. Participants wanted something like newest or most popular when the listings look similar.

Change: Add a sort dropdown next to the filters.

Priority: Could

Was it implemented? How? Not in this iteration. Filtering already covers most of the need. Leaving this as a future enhancement.